

Internship Report
Of
**Strategic Machine Learning Solutions for Agribusiness: Crop
Disease Detection and Deployment (Project AgriVision)**

Submitted
To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410030335

Name of Student: MIHIR SINGHAL

Class: 3CSE10

Batch: 2024-28

CANDIDATE'S DECLARATION

I, Mihir Singhal, do hereby declare that the internship report titled “Strategic Machine Learning Solutions for Agribusiness: Crop Disease Detection and Deployment (Project AgriVision)” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

mihir singhal

Student Name: Mihir Singhal

Roll No.: 2410030335

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank YuvaIntern and Mr. Kounal Gupta, Founder, YuvaIntern.com, for providing me the opportunity to undertake this internship as a Junior Machine Learning Engineer – Agribusiness, and for the guidance extended throughout its duration. I also thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature

Mihir Singhal

Student Name: Mihir Singhal

Roll No.: 2410030335

INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF EXPERIENCE

To Whomsoever It May Concern

This is to certify that **Mihir Singhal** successfully completed an internship with **YuvaIntern** in the role of **Junior Machine Learning Engineer - Agribusiness**.

Throughout the tenure of this internship, Mihir demonstrated consistent dedication, professionalism, and a strong willingness to learn and grow. Their contributions were valued by the entire team at YuvaIntern, and they conducted themselves with integrity and commitment throughout the program.

We commend Mihir's hard work and wish them continued success in all future endeavors.

Certificate No: YI/2026/208253/451552

Date of Issue: October 08, 2026

FOR AND ON BEHALF OF YUVAINTERNS

Kounal Gupta
Founder, [YuvaIntern.com](https://yuvaintern.com)

☐ **Henry Harvin America Head Office**
Henry Harvin Inc.
8 The Green, # 19614 Denver,
DE 19901, United States.

☐ **Henry Harvin Asia Pacific Office**
Henry Harvin India Education LLP
Henry Harvin House,
B-12, Sector-6, Noida (UP), India: 201301

☐ **Henry Harvin Middle East Office**
Henry Harvin Co. L.L.C.
2703, Blue Matrix, 27th floor The
Prime Tower, Business Bay, Dubai, UAE

CONTENTS

Chapter No.	Chapter Name	Page Nos.
1.	Candidate's Declaration	i
2.	Acknowledgement	ii
3.	Internship Completion Certificate	iii
4.	Project Description	
	4.1 Introduction	1
	4.2 Organization Profile	1
	4.3 Problem Statement	2
	4.4 Project Objectives	2
	4.5 Scope of the Project	3
	4.6 Technologies and Tools Used	3
	4.7 System Architecture	4
	4.8 Methodology	4
	4.9 Expected Outcomes	6
	4.10 Certificates of Completion and Communication Proof	6
5.	Bibliography/References	8

4. PROJECT DESCRIPTION

4.1 Introduction

This report documents the work undertaken during a four-week internship as a Junior Machine Learning Engineer – Agribusiness with YuvaIntern (in association with Henry Harvin Education and the National Skill Development Corporation), carried out from September 10, 2026 to October 08, 2026. The internship was designed to provide hands-on, project-based exposure to the application of machine learning in the agricultural technology (AgTech) sector, an area where data-driven systems can meaningfully improve outcomes for smallholder farmers and agribusiness operators alike.

Agriculture remains the economic backbone of many developing regions, yet crop pests and diseases continue to cause large, preventable yield losses — particularly among smallholder farmers who typically lack access to timely expert diagnosis. Advances in computer vision, deep learning, and the rapid growth of smartphone penetration in rural areas now make it feasible to put an accurate, near-instant diagnostic tool directly into a farmer's hands. Over the course of the internship, this opportunity was channelled into a structured, deliverable project, codenamed AgriVision, that progressed from strategic planning through data engineering, model development, and finally deployment and stakeholder communication planning.

The internship followed a phased, week-by-week structure: Week 1 focused on strategic project planning for an ML-based crop disease detection solution; Week 2 focused on data preprocessing and feature engineering for agribusiness datasets; Week 3 focused on model development and performance evaluation; and Week 4 focused on deployment strategy and stakeholder communication. Each week's work built directly on the previous week's deliverable, resulting in a coherent end-to-end plan for an agribusiness machine learning solution.

4.2 Organization Profile

YuvaIntern is an internship and upskilling platform that provides students and early-career learners with structured, project-based internship programs across various technology domains, including machine learning and data science. The program undertaken for this report was delivered in association with Henry Harvin Education, a professional training and upskilling organisation, and is recognised in collaboration with the National Skill Development Corporation (NSDC), a public-private initiative of the Government of India focused on skill development.

As part of this program, I was engaged in the role of Junior Machine Learning Engineer – Agribusiness, an internship position focused on developing and implementing

machine learning models to optimize agricultural processes and improve crop yield, in close (simulated) collaboration with agronomists and data scientists to analyze data, build predictive models, and provide insights for precision agriculture. The internship was carried out remotely over a duration of 4 weeks, from September 10, 2026 to October 08, 2026, and culminated in a Certificate of Experience bearing Certificate No. YI/2026/208253/451552, issued on October 08, 2026.

4.3 Problem Statement

Smallholder farmers currently detect crop disease primarily through visual inspection and word-of-mouth, and then travel to, or wait for, an agricultural extension officer to confirm the diagnosis. This process is slow, inconsistent across regions, and does not scale with the number of farmers who need support during an outbreak. By the time an official diagnosis and treatment plan arrive — often a week or more later — the disease has frequently progressed beyond the point where low-cost intervention is effective, forcing farmers into either yield loss or the over-application of agrochemicals as a precaution.

Machine learning, and image classification in particular, is well suited to this problem because visual disease symptoms such as leaf spots, discoloration, and wilting patterns can be reliably learned by trained models. A trained model can return a diagnosis in seconds rather than days, can be deployed at the edge or through a low-bandwidth server call to suit rural connectivity constraints, and — once trained — has a near-zero marginal cost per additional diagnosis, allowing the solution to scale to thousands of farmers without a proportional increase in expert staffing. Aggregated, anonymised diagnosis data can also support a live disease-outbreak map that informs early-warning systems for entire growing regions.

4.4 Project Objectives

The project objectives were framed using the SMART framework (Specific, Measurable, Achievable, Relevant, Time-bound) and scoped to what a pilot phase could realistically validate:

#	Objective	Target Metric
O1	Build an image classification model for the top 5 diseases affecting the pilot crop(s).	≥ 90% top-1 validation accuracy across target disease classes
O2	Deliver a mobile/USSD interface for photo-based diagnosis usable with intermittent connectivity.	Diagnosis response returned in under 15 seconds on a mid-range smartphone
O3	Reduce the average time between symptom onset and confirmed diagnosis for pilot farmers.	Cut average diagnosis turnaround from ~5–7 days to under 24 hours
O4	Validate real-world impact on yield loss and pesticide usage versus a control group.	Measurable reduction in disease-related yield loss in pilot plots
O5	Establish a data pipeline and feedback loop for continuous	Automated retraining pipeline live

#	Objective	Target Metric
	model improvement post-pilot.	before end of pilot

4.5 Scope of the Project

The project was scoped as a 16-week pilot plan (of which the 4-week internship covered the planning, data-strategy, model-design, and deployment-strategy stages) for an initial rollout in one or two crop value chains, with a clear path to scale across additional crops and regions in later phases. The intended target users of the resulting system are smallholder farmers, agricultural extension officers, and agribusiness cooperatives.

The scope of work delivered during the internship spanned four areas: (i) a strategic project plan defining the business problem, objectives, solution framework, timeline, resourcing, and risk register for the AgriVision crop-disease-detection initiative; (ii) a data preprocessing and feature engineering plan covering crop yield, weather, and soil quality datasets, supplemented by market and remote-sensing data; (iii) a model development and evaluation framework covering algorithm selection, evaluation metrics, and hyperparameter tuning strategy; and (iv) a deployment roadmap and stakeholder communication plan covering infrastructure, monitoring, and multi-audience messaging.

4.6 Technologies and Tools Used

The following technologies, tools, and techniques were planned and evaluated over the course of the internship:

- Programming and ML libraries: Python, scikit-learn, XGBoost/LightGBM, pandas, and standard MLOps practices for reproducible, auditable pipelines.
- Modelling approaches: Linear/Logistic Regression, Decision Trees, Random Forest, Gradient Boosting (XGBoost/LightGBM), Support Vector Machines, Convolutional Neural Networks (CNNs) with transfer learning from pretrained backbones (e.g., ResNet/EfficientNet), and time-series models (ARIMA/Prophet/LSTM).
- Explainability: SHAP-based feature attribution for tree-based models, and Grad-CAM overlays for CNN-based image classification.
- Data sources: crop yield records, meteorological/IoT weather data, soil quality (N, P, K, pH, organic matter) measurements, commodity market and price data, and remote-sensing vegetation indices (e.g., NDVI).
- Deployment infrastructure: containerized model hosting (Docker) on a managed cloud inference endpoint with autoscaling, a lightweight mobile/progressive web app with offline caching, an SMS/USSD fallback channel, and IoT sensor data ingestion via REST/JSON APIs and MQTT message queues.

- Experiment tracking and hyperparameter tuning: Grid Search, Random Search, and Bayesian Optimization (e.g., Optuna/scikit-optimize).

4.7 System Architecture

The proposed AgriVision solution was designed around a four-layer architecture: (1) a data layer that collects and curates labelled crop images from public agricultural datasets, partner cooperatives, and farmer-submitted photos; (2) a model layer built around a convolutional neural network or a lightweight vision transformer, fine-tuned via transfer learning from an existing image backbone; (3) an application layer consisting of a mobile app with offline capture and an SMS/USSD channel for feature-phone users; and (4) an insight layer that aggregates anonymised diagnoses into a regional dashboard for extension officers and agribusiness partners.

For deployment, a hybrid cloud–edge architecture was recommended over a purely cloud-centric or purely on-device design, given the mixed connectivity conditions typical of rural agribusiness settings:

Layer	Recommended Approach	Rationale
Model hosting (core)	Containerized model served via a managed cloud endpoint with autoscaling	Centralizes retraining, versioning, and monitoring; scales with seasonal demand spikes
Field access	Lightweight mobile app / progressive web app with offline caching	Field connectivity is unreliable; users must still see a result without live signal
Low-connectivity fallback	SMS/USSD or WhatsApp-based query-response	Reaches smallholder farmers without smartphones or data plans
Data ingestion	Scheduled batch sync from IoT sensors and manual field entries	Avoids requiring constant connectivity for data collection
Storage	Cloud data warehouse plus local on-device cache	Supports both retraining pipelines and offline resilience

4.8 Methodology

The internship was executed as four sequential weekly deliverables, each building on the previous week's output:

Week 1 – Strategic Planning for ML Solutions in Agribusiness

Defined the business problem, set SMART objectives, proposed the four-layer solution architecture described in Section 4.7, and laid out a phased 16-week roadmap covering data collection, model development, field testing, and deployment (Research & Scoping → Data Collection & Prep → Model Development → App Integration & Testing → Pilot Deployment & Review). Resourcing needs, a project risk register (covering data quality, field-condition generalisation, connectivity, farmer adoption,

diagnostic error, timeline slippage, budget overrun, and data privacy), and success-evaluation criteria were also documented.

Week 2 – Data Preprocessing and Feature Engineering

Produced a structured plan for cleaning and engineering features from three representative agribusiness datasets — crop yield records, weather data, and soil quality measurements — supplemented by market and remote-sensing data. This covered missing-value handling (time-aware interpolation, spatial/KNN imputation, multiple imputation for missing-not-at-random smallholder reports), outlier detection and treatment (region/crop-specific statistical thresholds, domain-based bounds, Isolation Forest/Local Outlier Factor for multivariate outliers), normalization and scaling strategies (min-max, z-score, log transforms, per-group scaling), and discretization/binning of variables such as rainfall categories and soil fertility index. Feature engineering strategies included growing degree days, rainfall distribution features, nutrient ratios, a soil fertility composite score, NDVI trajectory features, and a staged feature-selection process (filter → domain review → embedded selection → stability check).

Week 3 – Model Development and Performance Evaluation

Designed a complete model-development workflow: chronological/group-aware train-validation-test splitting to prevent data leakage across seasons and farms, a baseline-model requirement, k-fold cross-validation, and a staged model-selection process. Candidate algorithms were compared on interpretability, speed, and scalability, with Gradient Boosting (XGBoost/LightGBM) paired with SHAP explanations recommended as the primary approach for tabular yield/price prediction, and a CNN with transfer learning recommended for image-based disease detection. Evaluation metrics were defined for both classification tasks (accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix) and regression tasks (MAE, RMSE, MAPE, R^2), alongside domain-specific, business-impact metrics such as economic loss avoided and actionability lead time. Improvement strategies covered cross-validation techniques, hyperparameter tuning (Grid Search, Random Search, Bayesian Optimization), and mitigation of common failure modes such as overfitting on small datasets, class imbalance, and data drift.

Week 4 – Deployment Strategy and Stakeholder Communication

Developed a full deployment roadmap covering environment assessment, infrastructure setup (the hybrid cloud-edge architecture in Section 4.7), integration with existing systems (FMIS, ERP, IoT sensors, notification channels), and scalability planning (geographic scaling, seasonal load, model versioning, multi-tenancy). Post-deployment monitoring and feedback mechanisms were defined across model performance, data

drift, system health, adoption, and business impact, together with a retraining cycle. A stakeholder communication plan was built around five audiences — smallholder farmers, agronomists/extension officers, cooperative/farm managers, business/executive sponsors, and internal IT/operations staff — each with tailored messaging, formats, and technical depth, supported by a consolidated implementation checklist, risk-management table, and a 20-week rollout timeline.

4.9 Expected Outcomes

The plan developed during the internship is designed to generate the following outcomes once implemented and piloted:

- A working image classification model achieving $\geq 90\%$ top-1 validation accuracy across the top 5 target crop diseases for the pilot crop(s).
- A mobile/USSD diagnostic interface returning results in under 15 seconds on a mid-range smartphone, usable under intermittent connectivity.
- A reduction in average diagnosis turnaround time from approximately 5–7 days to under 24 hours for pilot farmers.
- A measurable, evidence-based reduction in disease-related yield loss and pesticide usage in pilot plots relative to a control group.
- An automated data pipeline and retraining/feedback loop supporting continuous model improvement beyond the pilot phase.
- A clear, evidence-based go/no-go recommendation, at the end of the 16-week pilot, on whether and how to scale the AgriVision solution to additional crops, regions, and farming communities.

Beyond the technical deliverables, the internship strengthened my practical understanding of applied machine learning project management, agribusiness domain constraints (seasonality, spatial variability, connectivity limitations), and the importance of designing ML systems that are both accurate and trusted by the non-technical stakeholders who ultimately rely on them.

4.10 Certificates of Completion and Other Communication Proof

The Certificate of Experience (Certificate No. YI/2026/208253/451552, dated October 08, 2026) issued by YuvaIntern on successful completion of the internship is attached in Chapter 3 – Internship Completion Certificate. The Internship Offer Letter, dated September 10, 2026, confirming the internship role, duration, and terms, is reproduced below as supporting communication proof.



INTERNSHIP OFFER LETTER

Date of Issue: September 10, 2026

Dear Mihir Singhal,

We are pleased to extend to you this formal offer for an Internship position as **Junior Machine Learning Engineer - Agribusiness** at **YuvaIntern**. We were impressed with your background, skills and enthusiasm, and we believe you will be a valuable addition to our team.

Internship Details

Start Date : September 10, 2026
End Date : October 08, 2026
Duration : 4 weeks
Location : Remote

Key Responsibilities:

As a **Junior Machine Learning Engineer - Agribusiness** intern, you will work on tasks and projects related to As a Junior Machine Learning Engineer in the Agribusiness sector, you will be responsible for developing and implementing machine learning models to optimize agricultural processes and improve crop yield. You will work closely with agronomists and data scientists to analyze data, build predictive models, and provide insights for precision agriculture..

Benefits & Learning Opportunities:

Throughout the duration of your internship, you will have the opportunity to:

- Develop practical skills in [relevant skills/tools].
- Gain hands on experience in [specific tasks/projects].

As part of joining our team, you may be required to sign a confidentiality or non-disclosure agreement. You are also expected to comply with all applicable company policies and regulations, which will be provided to you upon commencement.

We look forward to welcoming you to the team.



Kounal Gupta
Founder, [YuvalIntern.com](https://yuvalintern.com)

☐ **Henry Harvin America Office**
Henry Harvin Inc.
8 The Green, # 19014 Dover,
DE 19901, United States.

☐ **Henry Harvin Asia Pacific Office**
Henry Harvin India Education LLP
Henry Harvin House,
B-12, Sector-6, Noida (UP), India- 201301

☐ **Henry Harvin Middle East Office**
Henry Harvin Co. L.L.C.
2703, Blue Marlin, 27th floor The
Prime Tower, Business Bay, Dubai, UAE

5. BIBLIOGRAPHY/REFERENCES

- Frontiers in Artificial Intelligence (2023). A field-based recommender system for crop disease detection using machine learning.
- Dolatabadian, A. et al. (2025). Image-based crop disease detection using machine learning. Plant Pathology, Wiley Online Library.
- Mohanty, S. et al. (2016). Using Deep Learning for Image-Based Plant Disease Detection. Frontiers in Plant Science.
- ScienceDirect (2025). Precision agriculture in the age of AI: A systematic review of machine learning methods for crop disease detection.
- Ristaino, J. et al. (2021). The persistent threat of emerging plant disease pandemics to global food security, as referenced in crop-loss impact literature.
- Pedregosa, F. et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research.
- Chen, T. and Guestrin, C. XGBoost: A Scalable Tree Boosting System.
- Lundberg, S. and Lee, S. A Unified Approach to Interpreting Model Predictions (SHAP).
- YuvaIntern. Internship Offer Letter and Certificate of Experience, Certificate No. YI/2026/208253/451552 (2026).